

SQL Server 2014

Level 1: Introduction to SQL Server 2014

1.1. Overview of SQL Server

- What is SQL Server?
- SQL Server editions (Express, Standard, Enterprise)
- Features of SQL Server 2014
- SQL Server Architecture
- SQL Server Management Studio (SSMS)

1.2. Installing SQL Server 2014

- System requirements
- Installing SQL Server and SSMS
- Connecting to the server
- SQL Server services (Database Engine, Agent, etc.)

1.3. SQL Server Management Studio (SSMS)

- SSMS interface overview
 - Creating databases and tables using GUI
 - Exploring Object Explorer
 - Query Editor basics
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Level 2: Database Design & Creation

2.1. Database Fundamentals

- Database vs Table vs Record
- Relational Database Concepts (Primary key, Foreign key)
- ER Modeling Basics

2.2. Creating Databases & Tables

- `CREATE DATABASE, CREATE TABLE`
- Data types in SQL Server 2014
- Constraints: `PRIMARY KEY, FOREIGN KEY, NOT NULL, CHECK, DEFAULT, UNIQUE`

2.3. Data Types

- Numeric, Date/Time, Character, Binary
 - VARCHAR VS NVARCHAR
 - DATETIME, DATE, TIME, TEXT, BIT, etc.
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Level 3: Transact-SQL (T-SQL) Programming

3.1. Data Manipulation Language (DML)

- SELECT, INSERT, UPDATE, DELETE
- Filtering data with WHERE, IN, LIKE, BETWEEN
- Sorting with ORDER BY
- Aliasing columns

3.2. Functions and Expressions

- Built-in functions:
 - String: LEN(), SUBSTRING(), CHARINDEX()
 - Date: GETDATE(), DATEADD(), DATEDIFF()
 - Numeric: ROUND(), CEILING(), FLOOR()

3.3. Joins & Set Operations

- INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL JOIN
- UNION, INTERSECT, EXCEPT
- Using joins to combine data across tables

3.4. Aggregation and Grouping

- GROUP BY, HAVING
 - Aggregate functions: SUM(), AVG(), MAX(), MIN(), COUNT()
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Level 4: Advanced T-SQL

4.1. Subqueries and CTEs

- Scalar, Correlated, and Table Subqueries
- WITH clause and Common Table Expressions (CTE)

4.2. Stored Procedures and Functions

- `CREATE PROCEDURE`, `EXEC`
- `CREATE FUNCTION` (scalar and table-valued)
- Parameters in procedures

4.3. Triggers and Views

- `AFTER`, `INSTEAD OF` triggers
- `CREATE VIEW`, `UPDATE` through views
- Indexed views

4.4. Error Handling and Transactions

- `BEGIN TRAN`, `COMMIT`, `ROLLBACK`
 - `TRY...CATCH` blocks
 - Error functions: `ERROR_MESSAGE()`, `@@ERROR`
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Level 5: Database Administration (DBA)

5.1. Backup and Restore

- Types of backups: Full, Differential, Transaction Log
- `BACKUP DATABASE`, `RESTORE DATABASE`
- Recovery Models: Full, Simple, Bulk-logged

5.2. Indexing

- Clustered vs Non-Clustered Index
- `CREATE INDEX`, `DROP INDEX`
- Index fragmentation and Rebuilding

5.3. Security

- Logins and Users
- Roles and Permissions
- `GRANT`, `REVOKE`, `DENY`

5.4. SQL Server Agent & Jobs

- Creating scheduled jobs
- Alerts and Operators
- Monitoring job history

5.5. Monitoring & Performance Tuning

- Execution Plans
 - Query Analyzer
 - Index Tuning
 - DBCC commands (DBCC CHECKDB, DBCC SHOWCONTIG)
 - Extended Events and SQL Profiler
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Level 6: Business Intelligence Features (Optional)

6.1. SQL Server Integration Services (SSIS)

- ETL process overview
- Data Flow and Control Flow
- Connecting to different data sources

6.2. SQL Server Reporting Services (SSRS)

- Creating and publishing reports
 - Report parameters and expressions
 - Deployment and scheduling
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Level 7: Capstone Project / Practical Work

Project Ideas:

- **Library Management System**
- **Hospital/Clinic DBMS**
- **E-commerce database backend**
- **Student Results/Attendance System**
- Performance tuning on large tables
- Schedule automated reports